Project Design Phase-I
Solution Architecture

Date	07 March 2026
Team ID	MCA_DA2
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges the gap between business problems and technology solutions. It defines the structure, components, data flow, and technology choices for the project.

Architecture — 3-Tier System

TIER 1 — DATA: Consumption.csv → Python (validation) → MySQL 8.0
TIER 2 — BI / LOGIC: MySQL → Tableau Desktop → 12 Worksheets → 3 Dashboards + Story → Tableau Public
TIER 3 — PRESENTATION: Tableau Public embed URLs → Flask Web App → User Browser
SUPPORT: Git / GitHub | .env file (credentials) | Bootstrap 5 (CSS)

Components Table

S.No	Component	Description	Technology
1.	User Interface	Web dashboard in browser — embeds Tableau via iframe	HTML5, CSS3, Bootstrap 5
2.	Web Framework	Routes and renders HTML pages (Home, Dashboard, Story, About)	Python Flask 2.x,
3.	Business Intelligence	All visualisations, dashboards, story — connected to MySQL	Tableau Desktop / Tableau Public
4.	Database	Stores 16,600 rows — runs SQL aggregation queries	MySQL 8.0, MySQL Workbench
5.	Data Processing	Validates CSV, checks nulls, loads data to MySQL	Python 3.x, Pandas
6.	Visualisation Embed	Tableau dashboards embedded in Flask pages	Tableau Public iframe embed
7.	Version Control	Source code management and team collaboration	Git, GitHub